

Mohamed Ali Trabelsi

Full-Stack Software Engineer

Tunis, Tunisia | Open to relocation across Europe and international remote work
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PROFESSIONAL SUMMARY

Full-Stack Software Engineer specializing in TypeScript, NestJS and React Native, with professional experience since 2024 building web and mobile applications, REST APIs and backend services on PostgreSQL and MongoDB. Experienced with microservices, RabbitMQ, real-time features, third-party integrations, CI/CD and production deployment on cloud platforms. Additionally integrates LLM-based features into applications.

PROFESSIONAL EXPERIENCE

Full-Stack Engineer

Sep 2024 -- Present

IT Strategix

Tunis, Tunisia

- Design and develop backend services and REST APIs with NestJS and TypeScript for web and mobile application modules.
- Build full-stack features across React Native applications and backend services, integrating PostgreSQL and MongoDB data stores.
- Improved API and application performance, reducing response times by 35%.
- Contribute to application architecture and production deployment across multiple projects (project details confidential).
- Write automated tests with Jest and contribute to CI/CD pipelines in GitHub Actions.
- Work with cross-functional teams to translate business requirements into production-ready technical features.

Technologies: NestJS, TypeScript, React Native, Expo, PostgreSQL, MongoDB, Python, REST APIs, Jest, Git, GitHub Actions, CI/CD

Full-Stack Developer Intern

Jun 2025 -- Jul 2025

NAXXUM GROUP

Tunis, Tunisia

- Optimized an educational platform built with Node.js, Express, React and MongoDB, improving frontend performance and backend scalability.
- Enhanced recommendation features with AI-powered suggestions.
- Contributed to a better user experience and platform responsiveness.

Technologies: Node.js, Express.js, React.js, MongoDB, JavaScript, recommendation systems

Full-Stack Developer Intern

Jul 2024 -- Aug 2024

NAXXUM GROUP

Tunis, Tunisia

- Built and maintained features for an online teaching platform, including responsive interfaces and backend services.
- Implemented real-time student monitoring functionality.
- Strengthened application reliability through scalable feature development.

Technologies: React.js, Node.js, Express.js, MongoDB, REST APIs, real-time applications

End-of-Studies Intern, Full-Stack Development

Feb 2023 -- Jun 2023

SIGA

Tunis, Tunisia

- Designed and developed a task-tracking web application with Spring Boot, Angular and PostgreSQL.
- Built REST APIs and user management modules that improved internal coordination.

Technologies: Java, Spring Boot, Angular, PostgreSQL, REST APIs

EDUCATION

Engineering Degree in Software Engineering
ESPRIT

2023 -- 2026
Tunis, Tunisia

Bachelor's Degree in Computer Science
Faculty of Sciences of Tunis (FST)

2020 -- 2023
Tunis, Tunisia

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, Python

Backend & Architecture: NestJS, Node.js, Express.js, Spring Boot, REST APIs, Microservices, RabbitMQ, Real-time applications

Frontend & Mobile: React, React Native, Expo, Angular

Databases: PostgreSQL, MongoDB, Supabase

DevOps & Cloud: CI/CD, GitHub Actions, Docker, DigitalOcean, Vercel, Expo EAS, Firebase, Git

Testing: Jest

AI Integration: LLM integration, Generative AI, RAG, Prompt Engineering

PROJECTS

Most project source code is private (academic and company work). Walkthroughs available on request.

Smart Delivery Mobile App with AI Voice Assistant

- Built a cross-platform (iOS and Android) logistics application on a microservices architecture with NestJS and React Native, supporting three user roles from a single codebase.
- Implemented real-time delivery tracking with Google Maps and QR-code delivery confirmation; used RabbitMQ, Supabase and MongoDB in the backend.
- Integrated a voice assistant in Tunisian dialect (Darija) using speech-to-text, Gemini and OpenAI text-to-speech.
- Deployed the application and services with DigitalOcean and Expo EAS.

Stack: NestJS, React Native, RabbitMQ, Supabase, MongoDB, Google Maps, Gemini, OpenAI TTS, DigitalOcean

AI-Powered Education Platform

- Built a mobile education platform with Expo, React Native, NestJS and PostgreSQL.
- Integrated LLaMA to personalize tutoring and automatically match students with suitable teachers based on their needs.

Stack: Expo, React Native, NestJS, PostgreSQL, TypeScript, LLaMA

Travel & Social Activity Platform

- Designed and built a full-stack platform where travelers organize trips, share activities, find travel companions and connect with users based on destinations and interests.

Stack: Angular, NestJS, PostgreSQL, TypeScript, REST API

AI-Powered Fitness Coaching App

- Built a mobile fitness coaching app for beginners, integrating Gemini and Mistral AI to generate customized training programs based on user goals.

Stack: Swift, NestJS, MongoDB, Gemini, Mistral AI

LANGUAGES

Arabic -- Native; English -- C1; French -- B2

COMMUNITY INVOLVEMENT

Tunisian Red Crescent; IEEE SC Chapter -- FST Student Branch; Engineers Spark -- FST; Hult Prize ESSEC Tunis